Prayas JEE AIR (2025)

Advanced

WEEKLY TEST 25

DURATION : 180 Minutes | **DATE : 29/12/2024** | **M. MARKS : 180**

ANSWER KEY

PHYSICS		CHEMISTRY		MATHEMATICS	
1.	(4)	19.	(2)	37.	(9)
2.	(7)	20.	(6)	38.	(5)
3.	(5)	21.	(9)	39.	(3)
4.	(7)	22.	(3)	40.	(5)
5.	(1)	23.	(8)	41.	(2)
6.	(3)	24.	(7)	42.	(5)
7.	(2)	25.	(4)	43.	(2)
8.	(5)	26.	(5)	44.	(6)
9.	(A, B, D)	27.	(A, C, D)	45.	(B, D)
10.	(B, C)	28.	(A, B, C)	46.	(B, C)
11.	(C, D)	29.	(A, C, D)	47.	(B)
12.	(B, D)	30.	(B, D)	48.	(A, D)
13.	(A, C, D)	31.	(A, B, D)	49.	(A, C)
14.	(A, B, C)	32.	(A, B)	50.	(A, B)
15.	(B)	33.	(A)	51.	(A)
16.	(C)	34.	(C)	52.	(C)
17.	(D)	35.	(B)	53.	(C)
18.	(D)	36.	(A)	54.	(C)

PART-I (PHYSICS)

1. (4)

$$\frac{dm}{dt} \propto \pi r^2 \propto V$$

$$\Rightarrow \frac{dm}{dt} = k\left(\pi r^2 v\right)$$

$$\Rightarrow mg - \frac{v.dm}{dt} = \frac{mdv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = g - \frac{\pi k}{m} r^2 v^2$$

For $v = v_T, \frac{dv}{dt} = 0$

$$\Rightarrow g = \frac{\pi k}{m} r^2 v_T^2$$

$$\Rightarrow k = \frac{mg}{\pi r^2 v_T^2} \& \frac{4}{3}\pi r^3.\rho = m$$

$$\Rightarrow k = \frac{mg}{\pi v_T^2 \left\{\frac{3m}{4\pi\rho}\right\}^{2/3}} = \frac{g}{\pi v_T^2}\left[\frac{4\pi\rho\sqrt{m}}{3}\right]^{2/3}$$

$$= \frac{g}{\pi v_T^2}[100].\left[\frac{4\pi}{3}\right].\frac{1}{10}\left[\because m = \frac{4\pi}{3} grams\right]$$

$$= \frac{40}{3}\frac{g}{v_T^2} = \frac{4}{3}$$

2. (7)

Let h: height of gas at any time t.

$\Rightarrow$ Pressure of gas, P = $\rho g(H–h) + P_0$

Also, volume of gas, $V = Ah$

$$\Rightarrow \text{work} = \int PdV = \int_{H/3}^{H} \left[P_0 + \rho g\left(H - h\right)\right] Adh$$

$$= P_0 A\left[\frac{2H}{3}\right] + \rho gHA\left[\frac{2H}{3}\right]$$

$$-\frac{\rho gA}{2}\left[\frac{H^2 - H^2}{9}\right]$$

$$= \frac{2}{3}P_0 AH + \frac{2}{9}\rho gH^2 A$$

$$\Delta U = \left[\left(H^2 - x^2\right)\frac{3}{2}\rho gA\right]_{2H/3}^{0}$$

$$= \frac{2}{3}\rho gAH^2$$

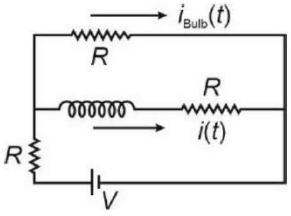
$$= \frac{6}{9}\rho gAH^2$$

3. (5)

$$\overrightarrow{Jerk} = \frac{d\vec{a}}{dt}$$

$$\Rightarrow \left|\overrightarrow{Jerk}\right| = \frac{v^2}{R}.\frac{d\theta}{dt} = \frac{v^3}{R^2} = \frac{10^3}{10^2} \text{ m/s}^3 = 10 \text{ m/s}^3$$

4. (7)



$$i(t) = \frac{V}{3R}\left[1 - e^{-3Rt/2L}\right]]$$

Also, $-\frac{Ldi}{dt} - iR + i_{Bulb}R = 0$

$$\Rightarrow i_{Bulb} = \frac{V}{6R}\left[2 + e^{-3Rt/2L}\right]$$

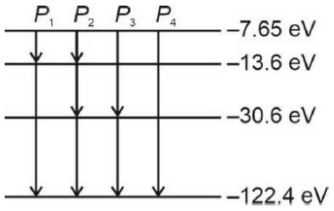
$$\Rightarrow \Delta t = \frac{2L}{3R} \text{ in } 2 = \frac{10^{-3}}{300} \times 0.69$$

5. (1)

Energy of nth state would be

$$E_n = \frac{-13.6(3)^2}{n^2} eV$$

$\Rightarrow$ Energy diagram would be:



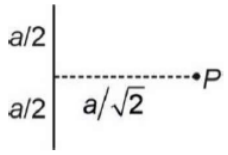
Only possibility P_1 and P_4 would give 1 photon each to eject an electron.

$$\Rightarrow \text{Number of photons} = \frac{2}{4} \times 10^8 = 0.5 \times 10^8$$

6. (3)

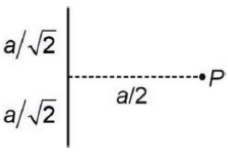
$\vec{B}$ due to body diagonal = $\vec{0}$

Due to edge:



$$\Rightarrow B = \frac{\mu_0 i}{4\pi\left(a/\sqrt{2}\right)}\left[\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right] \text{along } \frac{-\hat{j} - \hat{k}}{\sqrt{2}}$$

Due to face diagonal:



$$\Rightarrow B = \frac{\mu_0 i}{4\pi\left(a/2\right)}\left[\sqrt{\frac{2}{3}} + \sqrt{\frac{2}{3}}\right] \text{along } \frac{-\hat{j} - \hat{k}}{\sqrt{2}}$$

$$\Rightarrow \left|\vec{B}_{net}\right| = \frac{3}{2}\sqrt{\frac{2}{3}}\frac{\mu_0 i}{3\pi a}$$

7. (2)

Let $m \propto c^x G^y h^z$ or $m = Kc^x G^y h^z$

By substituting the dimension of each quantity in both sides

$$\left[M^1L^0T^0\right] = K\left[LT^{-1}\right]^x\left[M^{-1}L^3T^{-2}\right]^y\left[ML^2T^{-1}\right]^z$$

$$\left[M^1L^0T^0\right] = \left[M^{-y+z}L^{x+3y+2z}T^{-x-2y-z}\right]$$

By equating the power of M, L and T in both sides:

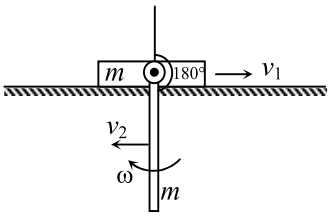
$-y+z=1, x+3y+2z=0, -x-2y-z=0$

By solving above three equations, we get

$x=\frac{1}{2}, y=-\frac{1}{2}$ and $z=\frac{1}{2}$

So, $m \propto c^{1/2}G^{-1/2}h^{1/2}$

8. (5)



There is no horizontal force, momentum is conserved

$mv_1 - mv_2 = 0$

For hinged point

$v_1 = \frac{L}{2}\omega - v_2$

Energy conservation,

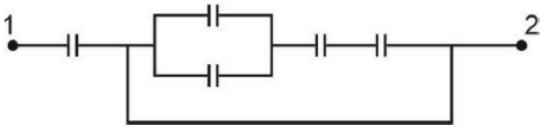
$$mgL = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}\frac{mL^2\omega^2}{12}$$

Solving, $v_1 = \sqrt{\frac{3}{5}gL}$, $v_1 = 10$ m/s.

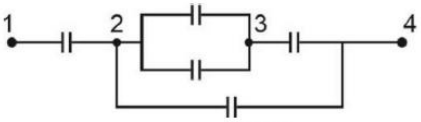
9. (A, B, D)

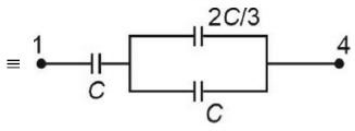
For points 1 & 2:

$\Rightarrow C_{eq} = C$ & four capacitors remain uncharged.



For points 1 & 4



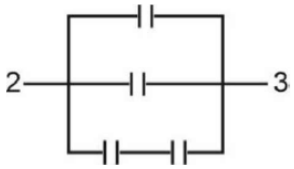


$$\Rightarrow C_{eq} = \frac{\frac{5C}{3}.C}{\frac{8C}{3}} = \frac{5C}{8}$$

$$\Rightarrow Q_{battery} = \frac{5CV}{8} \Rightarrow Q_{\min.} = \frac{\frac{2C}{3}}{\frac{2C}{3}+C} \times \frac{5CV}{8} \times \frac{1}{2}$$

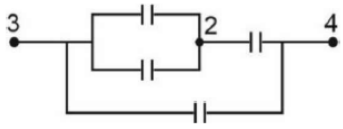
$$= \frac{2}{5} \times \frac{5CV}{8} \times \frac{1}{2} = \frac{CV}{8}$$

For points 2 & 3:



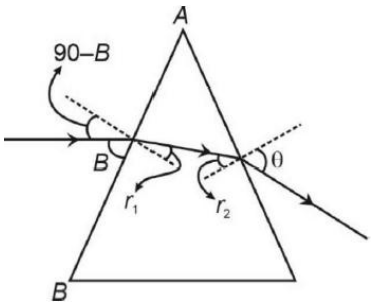
$$\Rightarrow C_{eq} = 2C + \frac{C}{2} = \frac{5C}{2}$$

For points 3 & 4



$$\Rightarrow C_{eq} = \frac{2C.C}{2C+C} + C = \frac{5C}{3}$$

10. (B, C)



1. $\sin(90 - B) = n.\sin r_1$

$\Rightarrow n \sin r_1 = \cos B$ .....(1)

Also, 1. $\sin\theta = n \sin r_2 = n \sin(A - r_1)$

$\Rightarrow \sin\theta = n\{\sin A \cos r_1 - \cos A \sin r_1\}$

$$= n\left\{\sin A.\sqrt{1-\frac{\cos^2 B}{n^2}} - \cos A.\frac{\cos B}{n}\right\}$$

$$= \sqrt{n^2 - \cos^2 B}.\sin A - \cos A\cos B \quad \ldots.(2)$$

For critical angle, $\theta \to 90^0$

$$\Rightarrow 1 = \sqrt{n^2 - \frac{3}{4}}.\frac{\sqrt{3}}{2} - \left(-\frac{1}{2}\right).\frac{\sqrt{3}}{2}$$

Deviation = $i_1 + i_2 - A = (90 - B) = 90 - 120 = 30^0$ As we can see from equation (2), increasing n increases $\theta \Rightarrow$ deviation increases. Also, r_1 decreases $\Rightarrow r_2$ increases $\Rightarrow r_2$ approaches θ_c.

11. (C, D)

The given case is of uniform circular motion. In which speed or kinetic energy remain constant. However, the direction of velocity and acceleration keep on changing although its magnitude remains constant.

12. (B, D)

For equilibrium, we have

Weight of block = Upthrust on the immersed part of the block

$\Rightarrow mg = (Ad)\rho g$

$\Rightarrow m = A\rho d$

$\Rightarrow d = \dfrac{m}{A\rho} = \dfrac{m}{\pi r^2 \rho}$

$\begin{pmatrix}\text{Restoring}\\ \text{force}\end{pmatrix} = \begin{pmatrix}\text{Upthrust experienced by the}\\ \text{additional part immersed}\end{pmatrix}$

$\Rightarrow F = -Ax\rho g$

$\Rightarrow m\ddot{x} = -Ax\rho g$

$\Rightarrow (A\sigma\ell)\ddot{x} = -Ax\rho g$

$\Rightarrow \ddot{x} + \dfrac{\rho g}{\sigma\ell}x = 0$

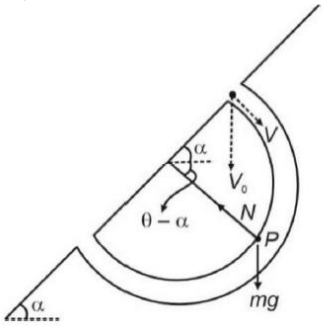
$\Rightarrow T = 2\pi\sqrt{\dfrac{\sigma\ell}{\rho g}}$

Since $m = (\pi r^2)\ell\sigma$

$\Rightarrow \ell\sigma = \dfrac{m}{\pi r^2}$

$\Rightarrow T = 2\pi\sqrt{\dfrac{m}{\pi r^2 \rho g}} = \dfrac{2\pi}{r}\sqrt{\dfrac{m}{\pi\rho g}}$

13. (A, C, D)



Since e = 0, $\Rightarrow$ only tangential component remains.

$\Rightarrow$ $V = v_0\cos\alpha$ is the initial speed. .....(1)

Let the ball be at point P at time t.

$\Rightarrow V_P = \sqrt{v^2 + 2g\{R\sin\alpha + R\sin(\theta-\alpha)\}}$ ...(2)

Also, $N - mg\sin(\theta-\alpha) = \dfrac{mv_P^2}{R}$

$\Rightarrow N = mg\sin(\theta-\alpha)$

$+\dfrac{m}{R}\left[2g(2R).\cos^2\alpha + 2gR\{\sin\alpha + \sin(\theta-\alpha)\}\right]$

$\Rightarrow N = mg\left[\sin(\theta-\alpha) + 1 + 2\sin\alpha + 2\sin(\theta-\alpha)\right]$

$N = mg\left[\sqrt{3} + 1 + 3\sin(\theta-\alpha)\right]$

$\Rightarrow$ N_{min} at $\theta = 0^0$ & N_{max} at $\theta = 150^0$

$N_{max} = mg = (4+\sqrt{3})mg$

For exit sped, we put $\theta = 180^0$ in equation (2).

$\Rightarrow V_{exit} = \sqrt{gR + 2gR\sqrt{3}} = \sqrt{(2\sqrt{3}+1)gR}$

14. (A, B, C)

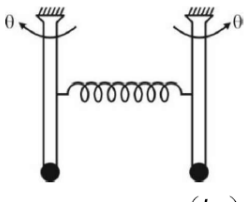
W_L: The rods oscillate independently

$\Rightarrow T = 2\pi\sqrt{\dfrac{l}{Mgd}}$

$\Rightarrow w = \sqrt{\dfrac{Mgd}{l}}$

$= \sqrt{\dfrac{2M.g.\frac{3L}{4}}{ML^2 + \frac{ML^2}{3}}} = \sqrt{\dfrac{9}{8}\dfrac{g}{L}}$

W_H:



Tension in the spring = $2.\left(\dfrac{L}{2}\theta\right) = L\theta$

$\Rightarrow$ Restoring torque on one rod

$= k(L\theta).\dfrac{L}{2} + Mg.\dfrac{L}{2}\theta + MgL\theta$

$= \left[\dfrac{kL^2}{2} + \dfrac{MgL}{2} + MgL\right]\theta$

$\Rightarrow -\left[\dfrac{kL^2}{2} + \dfrac{3MgL}{2}\right]\theta = l\alpha$

$\Rightarrow \left[\dfrac{ML^2}{3} + ML^2\right]\alpha = -[kL + 3Mg].\dfrac{L}{2}\theta$

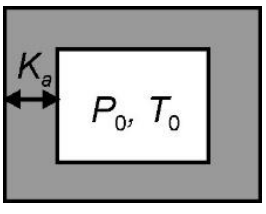
$\Rightarrow \alpha = -\dfrac{3}{8}\left[\dfrac{kL + 3Mg}{ML}\right]\theta$

$\Rightarrow \alpha = -\dfrac{3}{8}\left[\dfrac{k}{M} + \dfrac{3g}{L}\right]\theta$

$\Rightarrow W_H = \sqrt{\dfrac{3}{8}\left[\dfrac{k}{M} + \dfrac{3g}{L}\right]}$

15. (B)

$\dfrac{dQ}{dt} = c\dfrac{dT}{dt}$



$$\frac{dQ}{dt} = c\frac{dT}{dt} \quad \text{...(i)}$$

$$\frac{dQ}{dt} = \frac{kA}{t_0}\left(T - \frac{T_0}{2}\right) \quad \text{...(ii)}$$

From equation (i) and equation (ii)

$$\frac{dQ}{dt} = C\frac{dT}{dt} = -\frac{kA}{t_0}\left(T - \frac{T_0}{2}\right)$$

$$\Rightarrow \int_{T_0}^{T} \frac{C}{T - \frac{T_0}{2}} dT = \int_0^t -\frac{kA}{t_0} dt$$

$$\Rightarrow \quad T = \frac{T_0}{2}\left(1 + e^{\frac{-kA}{Ct_0}t}\right)$$

16. (C)

$kx = 3mg\sin\theta$

$kx = 3mg\sin\theta$

$$x = \frac{3mg\sin\theta}{k}$$

17. (D)

Fundamental frequency, $f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

T = force on dielectric slab.

To calculate force on slab, consider capacitor as combination of two capacitors one with slab and one with air.

$$F = -\frac{dU}{dx}; U = \frac{1}{2}\left[\frac{k\in_0 bx}{d} + \frac{\in_0 b(b-x)}{d}\right]V^2$$

$$\frac{dU}{dx} = \frac{1}{2}\left[\frac{k\in_0 b}{d} - \frac{\in_0 b}{d}\right]V^2 = \frac{1}{2}\frac{\in_0 bV^2}{d}(k-1)$$

Thus, $f = \frac{1}{2L}\sqrt{\frac{\in_0 bV^2(k-1)}{2d\mu}}$

18. (D)

Assume upper hemisphere,

$$B = \frac{2}{3}\pi R^3 \times \rho g$$

$F_2 = \pi R^2 \times \rho g(2R)$

$$\therefore \quad F_1 = F_2 - B = \frac{4}{3}\pi R^3 \rho g$$

PART-II (CHEMISTRY)

19. (2)

For s-subshell ($\ell = 0$)

m –1, 0, 1

∴ s subshell will have 3 orbitals and will accommodate 6 electrons.

Similarly,

p → 10 electrons

d → 14 electrons

Cr(24) : $1s^6\ 2s^6\ 2p^{10}\ 3s^2$

20. (6)

x = 2

y = 2

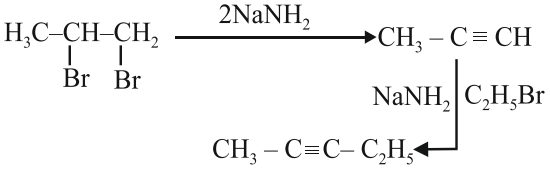
z = 2

So, (x + y + z) = 6

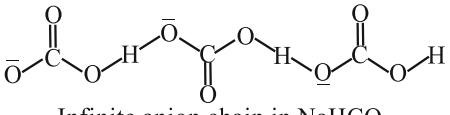
21. (9)

OF_2 has lower bond angle than H_2O.

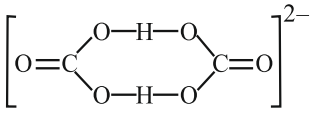
22. (3)



23. (8)

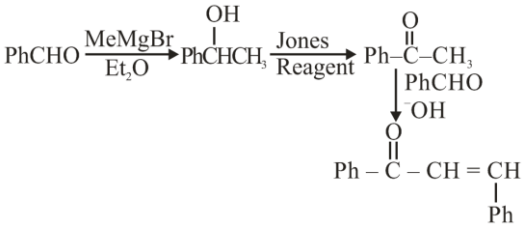


Infinite anion chain in $NaHCO_3$

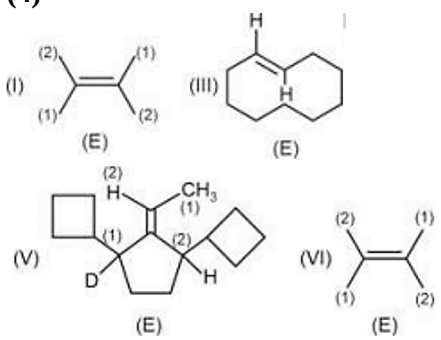


Dimeric anion in $KHCO_3$

24. (7)



25. (4)



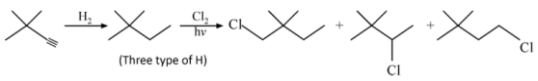
26. **(5)**

Sulphides of Hg^{2+}, Pb^{2+}, Bi^{3+}, Cu^{2+}, Cd^{2+} will not dissolve in yellow ammonium sulphide.

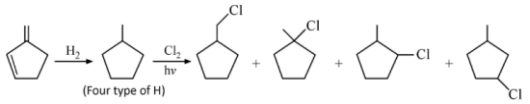
27. **(A, C, D)**

Compound Q possess three types of H atoms

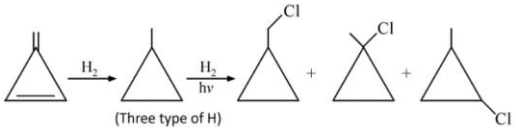
(A)



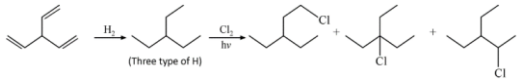
(B)



(C)



(D)



28. **(A, B, C)**

	PCl_5	$\rightarrow$	PCl_3	+	Cl_2
	(g)		(g)		(g)
$t=0$	a		0		0
$t=t_{eq}$	$a(1-\alpha)$		$a\alpha$		$a\alpha$

Total moles at equilibrium = $a + a\alpha$

$$P_{PCl_5} = \frac{a(1-\alpha)}{a(1+\alpha)} P_T$$

$$P_{PCl_3} = P_{Cl_2} = \frac{a\alpha}{a(1+\alpha)} P_T$$

$$P_T = 1 \text{ atm}$$

$$P_{PCl_5} = \left(\frac{1-\alpha}{1+\alpha}\right)$$

$$P_{PCl_3} = P_{Cl_2} = \frac{\alpha}{(1+\alpha)}$$

$$K_p = \frac{P_{Cl_2} \cdot P_{PCl_3}}{P_{PCl_5}} = \frac{\frac{\alpha}{1+\alpha} \cdot \frac{\alpha}{1+\alpha}}{\frac{1-\alpha}{1+\alpha}} = \frac{\alpha^2}{1-\alpha^2}$$

$$K_p = \frac{0.09}{1-0.09} = \frac{0.09}{0.91} = \frac{9}{91}$$

$$[a + a\alpha][M_{mix}] = aM_{PCl_5}$$

$$(1+\alpha)M_{mix} = M_{PCl_5}$$

$$M_{mix} = \frac{208.5}{1.3} \cong 160$$

$$\frac{r_{O_2}}{r_{mix}} = \sqrt{\frac{M_{mix}}{M_{O_2}}}$$

$$\frac{20/t}{V/t} = \sqrt{\frac{160}{32}}; \qquad \frac{20}{V} = \sqrt{5}$$

$$V = \frac{20}{\sqrt{5}} = 4\sqrt{5} = 8.95\text{ml}$$

29. **(A, C, D)**

$[Cr(H_2O)_6]^{+3}$

$t_{2g}^3 e_g^0$ C.F.S.E= $-1.2\ \Delta_0$

$[Fe(CN)_6]^{3-}$

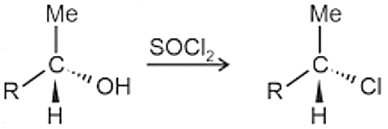
$t_{2g}^5 e_g^0$ C.F.S.E= $-2.0\ \Delta_0$

Ni^{2+} weak field or strong field

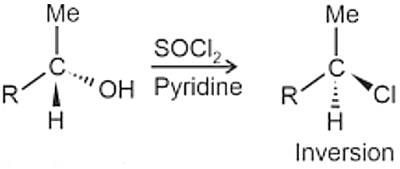
C.F.S.E = $-1.2\ \Delta_0$

30. **(B, D)**

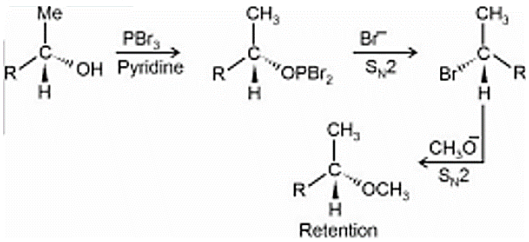
(A)



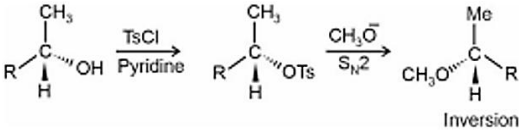
(B)



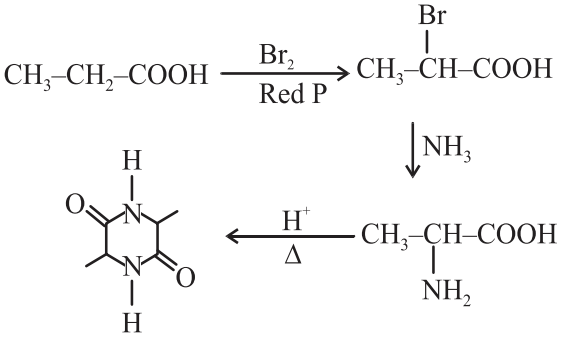
(C)



(D)



31. **(A, B, D)**



32. **(A, B)**

$2H_2O + Cu_2S \rightarrow 2Cu^{2+} + SO_2 + 4H^+ + 8e^-$

$2H_2O + Cus \rightarrow Cu^{2+} + SO_2 + 4H^+ + 6e^-$

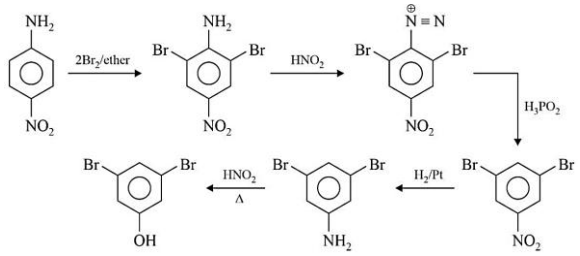
$5e^- + 8H^+ + MnO_4^- \rightarrow Mn^{+2} + 4H_2O$

n-factor, $Cu_2S = 8$

$CuS = 6$

$Ba(MnO_4)_2 = 2 \times 5 = 10$

33. (A)



34. (C)

K_{sp} of $Pb_3(AsO_4)_2 = 1.08 \times 10^{-28}$

$\Rightarrow 108\ S^5 = 108 \times 10^{-30}$

$S = 10^{-6}$ M

$R_{xn}[Pb_3(AsO_4)_2] + 6e^- \rightleftharpoons 3Pb + 2AsO_4^{-3}]$

$(K_{sp})Pb_3(AsO_4)_2] \rightleftharpoons 3Pb^{+2} + 2AsO_4^{-3}$

S 3S 2S

$\left[AsO_4^{-3}\right] = 2S = 2\times10^{-6}$ M

$Q_C = \left[AsO_4^{-3}\right]^2 = 4\times10^{-12}$

$\log Q_C = -12 + \log 4 = -12 + 0.60 = -11.40$

$E^\circ_{Cell} = -0.13 - \frac{0.06}{6}\log Q_C$

$= -0.13 - 0.01[-11.40]$

$= -0.130 + 0.114$

$\boxed{E^\circ_{cell} = -0.016 \text{volt}}$

35. (B)

$Q = nC\Delta T$

$C \rightarrow$ Molar heat capacity

$nC\Delta T = nC_v\Delta T +$ work

For PV^γ = constant

$$\text{Work done} = \frac{nR\Delta T}{1-\gamma}$$

Here, PV^{-1} = constant

So, $nC\Delta T = n\frac{3}{2}R\Delta T + \frac{nR\Delta T}{1-\gamma}$

$\Rightarrow C = \frac{3}{2}R + \frac{R}{1-(-1)} \quad \Rightarrow C = \frac{3R}{2} + \frac{R}{2} = 2R$

36. (A)

28% (w/w) ethylene glycol solution in water.

$\Rightarrow$ 28 grams of ethylene glycol in 100 gm solution

$\Rightarrow$ 28 grams of ethylene glycol in 72 gm solvent

Molality $\frac{28/M^\circ}{72\times10^{-3}} = \frac{28/62}{72\times10^{-3}}$

$= \frac{28\times1000}{72\times62} = 6.2724$ molal .

$\Delta T_f = K_f m = 1.86 \times 6.2724 = 11.667°C$

$(T_f)_{soln.} = -11.667°C$

Hence 28% (w/w) ethylene glycol solution is suitable for car radiator to prevent freezing at –11°C.

PART-III (MATHEMATICS)

37. (9)

Given,

Equation of hyperbola,

$$H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in \mathbb{N}$$

Now using the eccentricity formula, we get,

$$e^2 = 1 + \frac{3+n}{n+1}$$

$$\Rightarrow \quad e^2 = \frac{2n+4}{n+1} = \frac{2(n+2)}{n+1}$$

Now check when $(n + 1)$ is 9, 25, 49,.....is perfect square and then at the same tune $2(n + 2)$ should also be perfect square, (note here denominator is odd while numerator is even)

So checking for $n = 8 \rightarrow e^2 = \frac{20}{9}$

$$n = 24 \rightarrow e^2 = \frac{52}{25}$$

$$n = 48 \rightarrow e^2 = \frac{100}{49} \Rightarrow e = \frac{10}{7}$$

Hence, for $n = 48$ both $n + 1$ & $2(n+2)$ are perfect square.

Now we know that,

Length of latus rectum is given by

$$l = \frac{2b^2}{a} = \frac{2(n+3)}{\sqrt{n+1}} = \frac{2\times51}{7}$$

Hence, $21l = \frac{42\times51}{7} = 306$

38. (5)

Given,

A line *L* pass through the origin and be perpendicular to the lines

$L_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and

$L_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$

So, direction ratio of line will be,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 2 & 2 & 1 \end{vmatrix}$$

$= \hat{i}(-4) - \hat{j}(-5) + \hat{k}(-2)$

$= -4\hat{i} + 5\hat{j} - 2\hat{k}$

Hence, the equation of line *L* which passes through origin will be,

$\vec{r} = \sigma(-4\hat{i} + 5\hat{j} - 2\hat{k})$

Now finding the intersection point P we get,

$1 + \lambda = -4\sigma \ldots$ ....(1)

$-11 + 2\lambda = 5\sigma$ .....(2)

$-7 + 3\lambda = -2\sigma$ .....(3)

So, from equation (1) & (3) we get,

$1 + \lambda = -14 + 6\lambda \Rightarrow \lambda = 3, \sigma = -1$

Hence, point $P \equiv (4, -5, 2)$

Now finding the point Q which is foot of perpendicular from point $P(4, -5, 2)$ on line

$L_2 : (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k})$ we get,

Now any point on line L_2 will be

$Q((2\mu - 1)\hat{i} + 2\mu\hat{j} + (\mu + 1)\hat{k})$

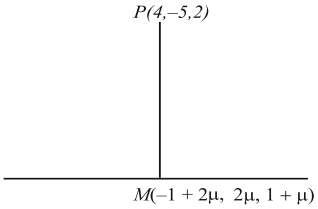
Now direction ratio of PQ will be

$(-5 + 2\mu)\hat{i} + (2\mu + 5)\hat{j} + (\mu - 1)\hat{k} = 0$

Now using the perpendicular condition we get,

$2(-5 + 2\mu) + 2(2\mu + 5) + 1(\mu - 1) = 0$

$\Rightarrow \mu = \frac{1}{9}$



Hence, $\alpha + \beta + \gamma = 2\mu - 1 + 2\mu + \mu + 1 = 5\mu = \frac{5}{9}$

$\Rightarrow 9(\alpha + \beta + \gamma) = 5$

39. (3)

$$I_2 = \int_0^1 \left(\frac{x}{5+x}\right)^{\frac{7}{2}} \cdot \left(\frac{1-x}{5+x}\right)^{\frac{9}{2}} \cdot \frac{dx}{(5+x)^2}$$

Put

$$\frac{x}{5+x} = t \Rightarrow \frac{5}{(5+x)^2}dx = dt \Rightarrow \frac{dx}{(5+x)^2} = \frac{1}{5}dt$$

$$\therefore I_2 = \int_0^{\frac{1}{6}} (t)^{\frac{7}{2}} (1-6t)^{\frac{9}{2}} \frac{dt}{(5)^{11/2}}; \text{Now put } 6t = \mu$$

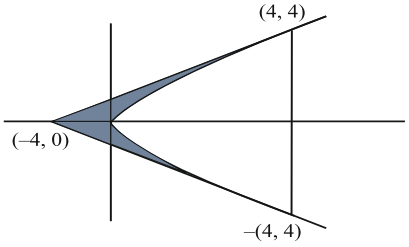
and simplify we get $I_2 = \frac{1}{5^{9/2} \times 6^{7/2}} I_1$ we

Conclude $a = 30$

40. (5)

$$\Delta = 2\int_0^4 \left(\frac{y^2}{4} - (2y - 4)\right) dy$$

$$= \frac{32}{3}$$



41. (2)

$$\lim_{n\to\infty}\left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \ldots + \frac{n}{(2n-1)^2}\right]$$

$$= \lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{n}{(n+r)^2} = \lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{n}{n^2 + 2nr + r^2}$$

$$= \lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{(r/n)^2 + 2(r/n) + 1}$$

$$= \int_0^1 \frac{dx}{(x+1)^2} = \left[\frac{-1}{(x+1)}\right]_0^1 = \frac{1}{2}$$

42. (5)

Let $x = y = 1$

$f(x) + f(y) + f(xy) = 2 + f(x).f(y)$

$3f(1) = 2 + (f(1))2$

$\Rightarrow f(1) = 1, 2$. But given that

$f(1) \neq 1$ so $f(1) = 2$

Now put $y = \frac{1}{x}$

$$f(x) + f\left(\frac{1}{x}\right) + f(1) = 2 + f(x) \cdot f\left(\frac{1}{x}\right)$$

$$\Rightarrow \quad f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$$

so $f(x) = \pm x^n + 1$

Now $f(4) = 17$

$\Rightarrow \pm (4)^n + 1 = 17 \Rightarrow n = 2$

$f(x) = +(x)^2 + 1$.

$\therefore \quad f(7) = 50$

43. (2)

$$\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} = \sqrt{1 + 1 + \frac{1}{4}} = \frac{3}{2}$$

$$\sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} = \sqrt{1 + \frac{1}{4} + \frac{1}{9}} = \sqrt{\frac{49}{36}} = \frac{7}{6}$$

$$\sqrt{1 + \frac{1}{3^2} + \frac{1}{4^2}} = \sqrt{1 + \frac{1}{9} + \frac{1}{16}} = \sqrt{\frac{169}{144}} = \frac{13}{12}$$

....

$$\sqrt{1 + \frac{1}{1999^2} + \frac{1}{2000^2}} = \frac{3998001}{3998000}$$

And now given expression,

$$\frac{3}{2} + \frac{7}{6} + \frac{13}{12} + \ldots + \frac{3998001}{3998000}$$

$$= \left(1 + \frac{1}{2}\right) + \left(1 + \frac{1}{6}\right) + \left(1 + \frac{1}{12}\right) + \ldots + \left(1 + \frac{1}{3998000}\right)$$

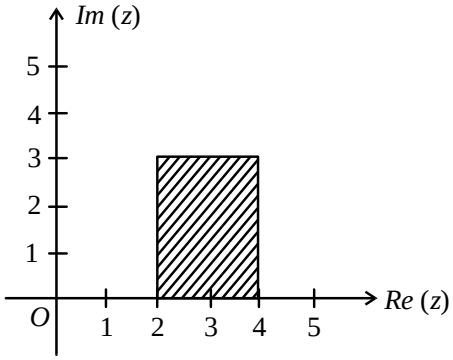
$$= 1999 + \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \ldots + \frac{1}{3998000}\right)$$

$$= 1999 + \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots +\right.$$

$$\left(\frac{1}{1999}-\frac{1}{2000}\right)\Bigg]=1999+\left(1-\frac{1}{2000}\right)$$

Hence, the value of n is 2000.

44. (6)



45. (B, D)

$a>0, b^2 \le 4ac$

$b>0, c^2 \le 4ab$

$c>0, a^2 \le 4bc$

$\Rightarrow a^2+b^2+c^2<4(ab+bc+ca)$
$[\because$ all equal is impossible$]$

Also $a^2+b^2+c^2>ab+bc+ca$
$(\because a,b,c$ are distinct$)$

$\Rightarrow \frac{a^2+b^2+c^2}{ab+bc+ca}\in(1,4)$

46. (B, C)

(A) $E_2(\underline{|125}) = \left[\frac{125}{2}\right]+\left[\frac{125}{2^2}\right]+\left[\frac{125}{2^3}\right]$

$+..+\left[\frac{125}{2^7}\right]=119$

$E_5(\underline{|125}) = \left[\frac{125}{5}\right]+\left[\frac{125}{5^2}\right]+\left[\frac{125}{5^3}\right]=31$

(B) $10^{10}-1$

(C) $\boxed{4}\square\square\square$

Case I : 2 alike $\rightarrow {}^3C_1 \times \frac{4}{\underline{|2}}$

Case II: 4 distinct $\rightarrow \underline{|4}$

$\boxed{5}\square\square\square$ Case I : 2 alike, 2 alike – $\frac{\underline{|4}}{\underline{|2}\underline{|2}}$

Case II: 2 alike, 2 distinct – 2 × $\frac{\underline{|4}}{\underline{|2}}$

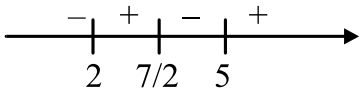
Total numbers = 60 + 30 = 90

(D) ${}^nC_2=5050$

47. (B)

$$\frac{1}{(x-2)(x-5)}(2x-7)>0$$

$$x+\left(2,\frac{7}{2}\right)\cup(5,\infty)$$



Limit $x\in\left(2,\frac{7}{2}\right)$ not in domain

48. (A, D)

Taking $x = r\cos\theta$ and $y = r\sin\theta$, so that $x^2 + y^2 = r^2$ and $y/x = \tan\theta$, we have $x\,dx + y\,dy = r\,dr$ and $x\,dy - y\,dx = x^2\sec^2\theta\,d\theta = r^2\,d\theta$

The given equation can be transformed into

$$\frac{r\,dr}{r^2\,d\theta}=\sqrt{\left(\frac{a^2-r^2}{r^2}\right)}\Rightarrow\frac{dr}{d\theta}$$

$$=\sqrt{\left(a^2-r^2\right)}\Rightarrow\frac{dr}{\sqrt{\left(a^2-r^2\right)}}=d\theta$$

Integrating both sides, then we get

$$\sin^{-1}\left(\frac{r}{a}\right)=\theta+c$$

or $$\sin^{-1}\left(\frac{\sqrt{\left(x^2+y^2\right)}}{a}\right)=\tan^{-1}\left(\frac{y}{x}\right)+c$$

or $$\sqrt{\left(x^2+y^2\right)}=a\sin\left\{\tan^{-1}\left(y/x\right)+c\right\}$$

or $$\sqrt{\left(x^2+y^2\right)}=a\sin\left\{\tan^{-1}\left(y/x\right)+\text{cosntant}\right\}$$

Also from equation (i)

$$\tan^{-1}\left(\frac{y}{x}\right)=\left\{\sin^{-1}\left(\frac{\sqrt{\left(x^2+y^2\right)}}{a}\right)-c\right\}$$

$$\text{or } y=x\tan\left\{\sin^{-1}\left(\frac{\sqrt{\left(x^2+y^2\right)}}{a}\right)-c\right\}$$

$$=x\tan\left\{\sin^{-1}\left(\frac{\sqrt{\left(x^2+y^2\right)}}{a}\right)+\text{ constant}\right\}$$

49. (A, C)

$$P\left(\frac{X}{Y}\right)=\frac{P(X\cap Y)}{P(Y)}=\frac{1}{2}\Rightarrow\frac{\frac{1}{6}}{P(Y)}=\frac{1}{2}$$

$$\Rightarrow P(Y)=\frac{1}{3}$$

$$P\left(\frac{Y}{X}\right)=\frac{P(Y\cap X)}{P(X)}=\frac{\frac{1}{6}}{P(X)}=\frac{1}{3}$$

$\Rightarrow P(X)=\frac{1}{2}$

$\therefore P(X\cup Y)=P(X)+P(Y)-P(X\cap Y)$

$=\frac{1}{2}+\frac{1}{3}-\frac{1}{6}=\frac{2}{3}$

$P(X\cap Y)=\frac{1}{6}, P(X)=\frac{1}{2}, P(Y)=\frac{1}{3}$

$\therefore P(X\cap Y)=P(X).P(Y)$

$\Rightarrow X$ and Y are independent

50. (A, B)

$M=\begin{bmatrix}0 & 1 & a\\ 1 & 2 & 3\\ 3 & b & 1\end{bmatrix}$ and adj $M=\begin{bmatrix}-1 & 1 & -1\\ 8 & -6 & 2\\ -5 & 3 & -1\end{bmatrix}$

$(\text{adj } M)_{11} = 2 - 3b$, $(\text{adj } M)_{22} = -3a$

$\Rightarrow 2 - 3b = -1$

$\Rightarrow b = 1$ and $-3a = -6 \Rightarrow a = 2$

51. (A)

In $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots.$

Put $x = i$ & $x = -i$, subtract & add we get

${}^nC_0 + {}^nC_1 - {}^nC_2 - {}^nC_3 + \ldots = 2^{n/2}$

$\left(\cos\frac{n\pi}{4}+i\sin\frac{n\pi}{4}\right)$

52. (C)

$\frac{h-4}{3a+b}=\frac{k-6}{-4a+b}$

$=\frac{-2((3a+b)4+(b-4a)6+(5a-3b))}{(3a+4b)^2+(b-4a)^2}$

$\therefore (h-1)^2 + (k-2)^2 = 25$

53. (C)

$1 + \cos^2(\alpha+\beta) - \sin^2(\alpha-\beta) - \cos 2\alpha \cos\beta$

$= 1+\cos2\alpha \cos2\beta - \cos 2\alpha \cos2\beta = 1$

54. (C)

1^∞ form

$\therefore e^{\lim\limits_{x\to\infty} x\left(\sin\frac{1}{x}+\cos\frac{1}{x}-1\right)} = e$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if